



Arsh Dhawan

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BS-MS

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EDUCATION

- **BS-MS**
Physical Sciences
IISER Bhopal, India
2023 - Present
- **Higher Secondary / +2**
Adarsh Jain, Delhi
2023
- **Secondary**
Scholars Rosary, Rohtak
2021

About

I am a third-year Physics Student interested in pursuing Theoretical Physics, especially interested in pursuing QFT and Mathematical Physics in future. I find both physics and mathematics fascinating especially topics like GR, QFT and Group Theory, and try to approach them by their most basic postulates. My experience under Prof. Ganguli has made me more inclined to understand the subject by focusing on essence of the subject, which translates better in Physics and Mathematics than in any other field. Moreover the opportunity under Dr. Naga Deepthi made me familiar with current advancements in Neutrino Physics while Dr. Rahul introduced me to Particle Physics and BSM Phenomenology which I briefly worked upon during Dec'25-Mar'26.

I am also familiar with a few specialized coding languages like GLoBES and SNoWGLoBES as well as proficient in Mathematica and MATLAB. Though, I am familiar I wish not to work in coding heavy projects and am more into mathematics/theory and phenomenology as well as model building. I am willing to do some coding for realization of these things and will most probably work alongside Arnav Kapoor for the same.

Experiences

- **Summer Internship 2026; Neutrino NSI Interactions and Sterile Neutrinos**
Under Dr. Deepthi, I am continuing my work on Scalar Non Standard Interactions and Neutrino Oscillations under Scalar NSI. We are hopeful for a publishable result and are almost near the end of this work.
Other than that I am working on understanding Sterile Neutrinos, one of the contenders of Dark Matter which shall have impact on my project(s) with Dr. Rahul Srivastava in near future, if not MS thesis.
Dr. Deepthi K. N.
- **Project: Discrete Symmetries and Generating Neutrino Masses (SeeSaw Mechanism) by Discrete Symmetry Extensions of Standard Model**
Under Dr. Arnab Chaudhury and co-supervisor Dr. Behera, I understood the Discrete Symmetry Groups and their importance in Particle Physics. I primarily focused on S_3 , A_4 and A_5 groups. We will then use one of these groups to extend Standard Model using Modular Symmetries and using that to explain Neutrino Masses using (Type-I, Type-II, Type-III) Seesaw Mechanism.
Dr. Arnab Chaudhury and Dr. Mitesh Behera
- **Non-Hermitian Dynamics in 2-flavour Neutrino Oscillations (Summer 2026)**
Under Dr. Poonam Mehta I am working on a project which as planned would attempt in studying Non Hermitian Physics starting from OQS, and better understanding Entanglement and Bell Theorem.
Following this we shall attempt to use G-metric and Density Matrix approach in Non-Hermitian Physics for 2-flavour neutrino-oscillations. This project is heavily influenced by Dr. Poonam's recent paper on
Dr. Poonam Mehta
- **Project: Dark Matter and Its Contenders (2025-26)**
During winter (Dec-Mar'26), I took up a project under Dr. Rahul Srivastava, Associate Professor, IISER Bhopal where I approached Dark Matter Phenomenology using Particle Physics (BSM). For this I referred to Dr. Gianfranco Bertone white paper and book on the same alongside other resources. Other than this I also worked on limiting cases of neutrino physics under constraints imposed by solar, atmospheric and Supernova neutrinos.
Dr. Rahul Srivastava

ICTP offers a select few students, courses in a few topics every year. During Sep’25-Feb’26, I was part of one such course focused upon "Riemannian and Complex Geometry" which introduced me to topology, differential, riemannian and complex geometry along with some concepts/practical applications in General Relativity.

• Teaching Assistant, Quantum Physics (2025)

Dr. Dr.Nirmal Ganguli

Under Prof. Ganguli, I acted as TA for Quantum Physics course, helping other students understand the topics better by solving their doubts and sometimes, teaching a few during the Tutorials.

• Summer Internship 2025; Supernovae Neutrinos and Neutrino Oscillations

Dr. N. K. Deepthi

Under Dr. Naga Deepthi, I worked in the field of Neutrino Oscillations covering the MSW effect in matter. Later, I focused on the study of supernova and the neutrino emission during such astrophysical events. I was also introduced to the fundamental concepts of Open Quantum Systems as well as basic QFT and Particle Physics, required for the project. I also delved into Computational aspect, making use of GLoBES to understand neutrino phenomenology better. Later following the collaboration, I studied Amol Dighe’s paper on ‘Identifying FLux from Supernova Explosions’ as well as Kuo and Mikheyev’s paper on Resonance in Neutrino Oscillations.

• Reading Project; Quantum Mechanics

Dr. Arnab Rudra

Under Dr. Rudra, in an extended program, he encouraged learning quantum mechanics by a problem solving approach, making us familiar with basics of QM as well as dirac notation.

• Summer School; General Relativity and Cosmology

Dr. Shibesh Kumar Jas, Director, PICS

Under Dr. Shibesh, I got familiar with General Relativity and Black Hole Thermodynamics, using computational Tools like Mathematica throughout this project.

Working Paper

• NSI Scalar Interaction in Neutrino Physics
I am currently working on a paper with Chinmay Bera(Ph.D) and Dr. Deepthi N. K. on finding mixing matrices as well as effective masses when considered Vacuum, matter and NSI Scalar effects on Neutrino Oscillation simultaneously. It is a continuation of a paper by the latter 2 with Prof. R. Mohanta. You can find the methodology and updates here.

TECHNICAL SKILLS

- Linux
- C
- MATLAB
- Mathematica
- LaTeX
- GLoBES
- SNowGLoBES

for Neutrino Physics

Relevant Courses

- Mechanics
- B+
- Quantum Physics
- A
- Calculus
- B

• Groups and Symmetries B • Quantum Mechanics-I B • Standard Model of Particle Physics A	[Audited]
Knowledge from other Sources	
• Relativistic QM • Basic QFT • Open Quantum Systems	
Extra-Curricular Activities	
• Playwright AARAMBH [IISER-B Drama Club] • Head Writer AARAMBH[IISER-B Drama Club] • Content writer Singularity Tech Fest [IISER-B] • Content writer Mahuaa Fest [IISER-B]	2023-24 2024-2025 2023-24 2023-24
Volunteer	
• Literary Fest, Arts and Letters • Integration Bee Acted as event organizer of "Integration Bee" for IISER-B Math Fest, Continuum.	2024 2024